

MODEL QUESTION PAPER

TED (15)5092

Reg No.....

REVISION 2015

Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

PLASTIC TECHNOLOGY

Maximum marks 100

Time 3 hours

(PART -A)

(Answer the following question in one or two sentences. Each question carries 2marks)

- I
1. Classify the plastics according to their thermal response with examples?
 2. Write the catalyst or initiator used for the preparation of HDPE,PS,LDPE & isotactic PP?
 - 3 Define thermoplastic polyesters and thermo set polyesters?
 4. Distinguish between Novolac and resol PF resins?
 5. What is coupling agents? Give one example? (5X2=10)

(PART -)

(Answer any FIVE questions. Each question carries 6 marks)

- II
1. State the importance of plastic in modern life. Write its advantages and disadvantages with other materials?
 2. List the different type of polyethylene and write the properties and application of each?
 - 3 What are the properties and application of Shellac and Bitumen?
 4. Explain the monomer preparation of caprolactam and Hexa Methylene Diamine(HMDA)?
 5. What are the properties and application of poly Urethane?
 6. Explain the curing reactions of Novolac and Epoxy resins?
 7. Define plastic compounding and list the various compounding ingredients with example? (5x6=30)

PART -C

(Answer any one full question from each unit.Each full question carries 15 marks)

MODULE- I

III a) Explain the manufacture and application of Toughened and Expanded polystyrene (8)

b) Explain the manufacture and applications of PVC by suspension polymerisation technique with a formulation? (7)

OR

IV a) Explain the manufacture and properties of poly propylene with a flow diagram? (8)

b) Explain the manufacture of PMMA by mass polymerisation technique? Give its properties (7)

MODULE- II

V a) Describe the production of Nylon 6, 6 starting from its monomer preparation. (8)

b) Write the structure, properties and applications of Teflon. (7)

OR

VI a) Explain the monomer preparation, polymerization properties and application of polycarbonate (8)

b) Define thermoplastic polyester and thermo set polyesters with examples? (7)

MODULE- III

VII a) Describe industrial production, properties and applications of PF resin (8)

b) Explain the monomer preparation and polymerization of silicone plastics (7)

OR

VIII a) Describe the raw material, industrial production, and applications of MF resin

b) Explain the monomer preparation and polymerization of epoxy resin? (7)

MODULE- IV

IX a) Name the different types of Antidegradents for plastic products?Explain the action of antioxidant with a typical example? (8)

b) Explain different types of plastic compounding equipment? (7)

OR

X a) Explain the changes in fillers in PF molding powder formulations? (8)

b) Explain the action of crosslinking agent in UP resin (7)

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